EYEQ: A Hybrid Quantum–Classical Eye-Gaze Typing System on Commodity Webcams

Devash Sivakumar
Dept. of Computer Science and Engineering
Vels Institute of Science, Technology
and Advanced Studies
Chennai, India
23619110@vistas.ac.in

R. Anandan
Dept. of Computer Science and Engineering
Vels Institute of Science, Technology
and Advanced Studies
Chennai, India
anandan.se@vistas.ac.in

V. Maruthi
Dept. of Computer Science and Engineering
National Institute of Technology
Tiruchirappalli
Tiruchirappalli, India
maruthiphd.83@gmail.com

***Abstract*—Eye typing on a commodity webcam faces two problems in series, and they are not the same kind of problem: estimating a screen coordinate from a low-resolution face, and committing a character from a noisy coordinate and a language context. We address both. For the decision problem we introduce a *quantum conditional selection rule*: candidate keys occupy a five-qubit register with amplitudes carrying the gaze likelihood, a language-conditioned diagonal unitary attaches both magnitude and phase, a consistency oracle admits only hypotheses that re-encode to what was gazed, and the Born rule commits. We prove it contains the deployed log-linear fusion exactly at zero phase and that its interference cross-terms are unreachable classically. Measured on identical noise at matched prompt rate it is worth a median -0.03 percentage points, and we give the mechanism: coherence is anti-correlated with correctness. For the estimation problem we replace eyelid-relative normalisation of the iris offset with a canthus-relative one, cutting held-out error from 5.55° to 3.76° across eleven sessions from nine participants, improving every participant, and roll-invariant by construction.**

***Index Terms*—Quantum conditional selection rule; interference-based fusion; amplitude amplification; eye-gaze typing; webcam gaze estimation; dwell selection; assistive text entry; facial landmarks; calibration.**

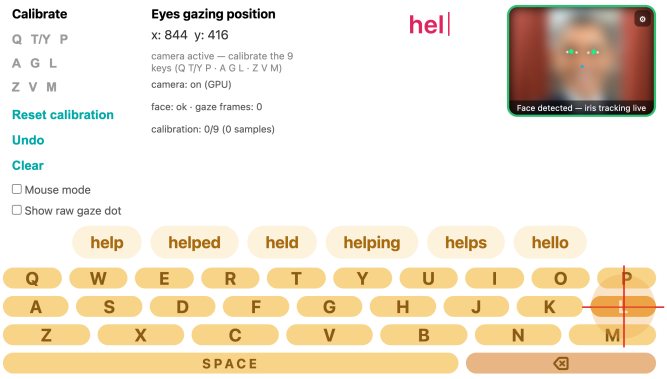


Fig. 1. The deployed interface, captured live. In the preview panel the green markers are the iris centres and the white markers beside them are the medial and lateral canthi — the landmarks (3) is built on. The user has gaze-typed `hel`; the crosshair rests on L, which is filling its dwell, and the language model offers six completions. The camera image is blurred for privacy; the tracking overlay is a separate canvas and is unaffected.

I. Introduction

Eye typing gives people with severe motor impairment a writing channel that needs no limb movement. Infrared trackers reach 0.25–0.5° of gaze error but cost more than the computers they attach to, so a large literature pursues the same capability with a consumer webcam alone. Fig. 1 shows the system this paper reports.

That literature reports *classifiers*: the fraction of frames assigned to the correct region, over two to nine regions, with figures of 85–$93\,\%$ common. That is a fair measure of a fair design, since such systems select by blinking or scanning and need only a coarse direction, but it is not comparable to continuous angular error and cannot tell a reader whether a user could point at a key. We take the harder formulation throughout — a continuous coordinate, held-out targets, degrees of visual angle — and working in that regime exposed a defect in a step the literature treats as settled.

The decision problem received the more ambitious treatment and returned less. Gaze keyboards multiply a gaze posterior by a language prior, a product of *marginals*; a marginal has already discarded whether a letter's support came from one confident word or fifty diffuse ones. Our rule keeps the hypotheses distinct, places them in superposition with gaze-derived amplitudes, and attaches a *phase* to each so that context-coherent hypotheses interfere. We prove no classical mixture can express that, measure it, and find it worth nothing — for a reason sharp enough to be the more durable contribution.

TABLE I
REPRESENTATIVE WEBCAM GAZE-KEYBOARD SYSTEMS. THE RIGHTMOST COLUMN IS THE POINT OF THE TABLE: REPORTED FIGURES MEASURE REGION CLASSIFICATION, NOT CONTINUOUS GAZE ERROR, AND ARE THEREFORE NOT COMPARABLE WITH THIS WORK.

Work	**Iris localisation**	**Head pose**	**Reported**
Waideman *et al.* [1]	4-stream CNN (face, eyes, head)	**Yes**, 4th channel	93 % L/R, **76 % U/D**
Islam *et al.* [2]	dlib 68-pt, midpoint	Movement prohibited	CPM/WPM only
Rafid *et al.* [4]	**MediaPipe iris**	No	14.53 CPM
Kathpal *et al.* [7]	Horizontal/vertical gaze ratio	No	91 % / 85 % (glasses)
Padilla *et al.* [9]	Sobel + thinning + Circular Hough	No	FPGA ≡ MATLAB
Li *et al.* [10]	SMI ETG 2w (commercial IR)	Noted, unsolved	—
This work	**MediaPipe iris, canthus-normalised**	**Implicit (canthus frame)**	$\mathbf{3.76^\circ}$ **continuous**

A. Contributions

1) A quantum conditional selection rule for gaze–language fusion, with a containment proposition showing it recovers the deployed log-linear fusion exactly at zero phase and an expressiveness proposition showing its cross-terms are unreachable classically.
2) A measured null with a diagnosed mechanism. Interference is worth a median -0.03 percentage points at matched prompt rate because coherence is *anti*-correlated with correctness, and no retuning fixes a sign.
3) Canthus-relative normalisation, which cuts held-out error by $32.3\,\%$ across nine participants, subsumes explicit head pose, is roll-invariant by construction to $\sim 10^{-15}$, and explains an asymmetry reported but undiagnosed in prior work ($93\,\%$ horizontal against $76\,\%$ vertical).

II. RELATED WORK AND POSITIONING

Table I groups representative webcam gaze keyboards by how the iris is localised, because that choice determines what the rest of the system can do. Threshold-and-projection methods [8] are inexpensive but need a per-illumination threshold. Circle fitting [9] is the only surveyed work targeting dedicated hardware. The largest group [2], [3], [5], [7] uses the 68-point *dlib* predictor and derives an eye aspect ratio; that model contains *no iris landmarks at all*, which is why all four select keys by blinking or scanning rather than by pointing. Only [4] and [6] adopt the dedicated iris ring of MediaPipe [11], the prerequisite for continuous regression. Enlarging targets and recovering the word with a language model is the established trade in text entry [15]; our addition is the constraint direction, since the oracle of Section IV admits only hypotheses that re-encode to exactly the gazed sequence.

Two facts in Table I motivate this work. First, **no surveyed work reports gaze error in degrees**, so our results cannot be ranked against them and we do not attempt it. Second, **head pose is almost entirely absent**: one work models it implicitly [1], one *prohibits* head movement and reports the discomfort as a limitation [2], and one notes that head shaking causes error without addressing it [10]. Section VI explains why that omission has cost less than expected.

III. FRONT END: ESTIMATING THE COORDINATE

A. The vertical-asymmetry observation

The observation that motivates our method is already in the literature but is not commented upon there. Waideman and Aquino Junior [1] report directional accuracy split by axis, $93\,\%$ for left/right against $76\,\%$ for up/down, and do not explain the gap. Consider how the vertical channel is conventionally formed. With $\mathbf{p}$ the iris centre and t, b the upper and lower eyelid landmarks, the standard feature is

$$v = \frac{p_y - t_y}{b_y - t_y + \epsilon}, \tag{1}$$

the iris height as a fraction of the *eyelid aperture*. That denominator is not a fixed anatomical quantity: it contracts with every squint, every partial blink, and with ambient brightness, so v changes without the eye having moved. The horizontal channel, normalised by the distance between the eye corners, suffers no equivalent contamination. An eyelid-normalised vertical channel should therefore be measurably worse than its horizontal counterpart, which is what [1] reports. Our method follows directly: normalise against a structure that does not move.

B. Canthus-relative normalisation

Let $\mathbf{i}$, $\mathbf{m}$ and $\mathbf{l}$ be the iris centre and the medial and lateral canthi of one eye. We define the eye axis and its perpendicular,

$$\mathbf{u} = \frac{\mathbf{m} - \mathbf{l}}{\|\mathbf{m} - \mathbf{l}\|}, \qquad \mathbf{u}^{\perp} = \begin{bmatrix} -u_y \\ u_x \end{bmatrix}, \tag{2}$$

take the offset from the canthus midpoint as $\mathbf{d} = \mathbf{i} - \frac{1}{2}(\mathbf{m}+\mathbf{l})$, and express it in the eye's own frame, scaled by the inter-canthal distance $w = \|\mathbf{m} - \mathbf{l}\|$:

$$c_{\parallel} = \frac{\mathbf{d} \cdot \mathbf{u}}{w}, \qquad c_{\perp} = \frac{\mathbf{d} \cdot \mathbf{u}^{\perp}}{w}. \tag{3}$$

Two properties follow. The scale w is a Euclidean distance rather than a coordinate difference, so it does not collapse as the head turns. And because $\mathbf{u}$ rotates with the head, (3) is invariant to head roll *by construction* — no roll angle is estimated, and none is needed. Fig. 2 contrasts the two normalisations.

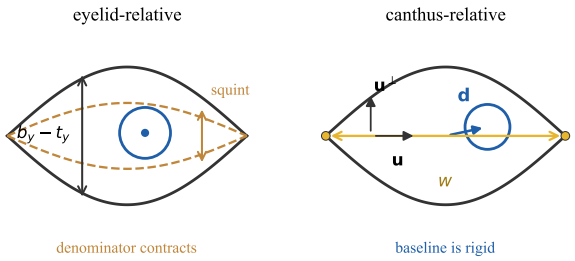


Fig. 2. Eyelid-relative against canthus-relative normalisation of the iris offset. Only the latter is invariant to eyelid aperture and to head roll.

C. Feature vector and calibration

The full vector concatenates the four conventional ratios, head translation and scale proxies, a bias term, and the four canthus components, giving $d = 12$. Screen coordinates come from ridge regression, $\mathbf{W} = (\mathbf{F}^\top\mathbf{F} + \lambda\mathbf{I})^{-1}\mathbf{F}^\top\mathbf{T}$, fitted to a nine-target per-user calibration of five fixations each. Predictions pass through a One Euro filter [12], and dwell hysteresis retains the target until the estimate leaves the key by a margin, which converts boundary jitter into a stable selection.

IV. A Quantum Conditional Selection Rule

A. The rule

At commit time the deployed system scores each candidate key k as

$$\text{score}(k) = P(k \mid \text{gaze}) \cdot \max\big(P(k \mid \text{prefix}),\, \varepsilon\big)^{\beta}, \quad (4)$$

with $\beta < 1$ damping the prior so the language model advises rather than vetoes. The second factor is a marginal: the total frequency mass of every vocabulary word continuing the prefix with k. Forming that sum destroys the distinction between a letter backed by one dominant continuation and one backed by a diffuse spray agreeing on nothing else.

Our rule keeps the hypotheses. Let H be the vocabulary completions of the current prefix, each with weight w_h and proposing next letter k_h. Candidates occupy a five-qubit register spanning a–z and a word boundary, prepared as $|\psi\rangle = \sum_k \sqrt{P(k \mid \text{gaze})}\,|k\rangle$. Writing $v_h = w_h / P(k \mid \text{prefix})$ for the within-letter weights, so $\sum_h v_h = 1$, a language-conditioned diagonal unitary gives

$$a(k) = \sqrt{P(k \mid \text{gaze})}\; P(k \mid \text{prefix})^{\beta/2} \underbrace{\textstyle\sum_{h:\,k_h = k} v_h\, e^{i\varphi_h}}_{c(k)}, \quad (5)$$

and the Born rule commits $\arg\max_k |a(k)|^2$. The factor $c(k)$ is the *coherence* of letter k, a weighted mean phasor whose squared modulus $\sum_h \sum_{h'} v_h v_{h'} \cos(\varphi_h - \varphi_{h'})$ carries the interference in its off-diagonal terms. Since $|c(k)| \le 1$ with equality exactly when the supporting hypotheses agree in phase, coherence penalises letters reachable only through mutually incoherent continuations. Phases come from each hypothesis's mean per-character log-frequency, rescaled to $[0, 1]$ over H and multiplied by a scale $\sigma_\varphi \pi$.

B. Containment and expressiveness

Proposition 1 (Containment): At $\sigma_\varphi = 0$, (5) reduces to (4). *Proof.* With all $\varphi_h = 0$, $c(k) = \sum_h v_h = 1$ for every k, so $|a(k)|^2 = P(k \mid \text{gaze})\, P(k \mid \text{prefix})^{\beta}$, which is (4) up to the shared normalisation. □

This is not a formality. A model class that cannot express the incumbent cannot be said to have beaten it, so the comparison in Section VI is meaningful only because the baseline lies strictly inside the family being tested. We assert the identity numerically to 10^{-12} over randomised inputs rather than relying on the derivation.

Proposition 2 (Expressiveness): No classical mixture over the same hypotheses reproduces the cross-terms of (5). *Proof sketch.* A mixture assigns letter k probability $\sum_h v_h^2$ after normalisation — the diagonal alone. The off-diagonal terms depend on phase differences, and a mixture has no phase. Since $(\sum_h v_h)^2 > \sum_h v_h^2$ whenever more than one hypothesis contributes, the two disagree. □

Concretely, two hypothesis sets with *identical* marginals, and therefore indistinguishable to (4), receive different posteriors under (5), and the diffuse set is the one penalised.

C. Oracle, amplification and cost

The cluster decoder's correctness guarantee is a predicate: a hypothesis is admissible only if re-encoding it reproduces exactly the cluster sequence gazed. As an operator this is a projector O_S, and it is what prevents the language model from ever proposing a word the gaze did not support. Given a projector, amplitude amplification [13], the generalisation of Grover search [14] to a non-uniform start, rotates amplitude into the marked subspace in $O(1/\sqrt{p_S})$ steps with $m = \lfloor (\pi/4)/\arcsin\sqrt{p_S} \rfloor$ rounds. The intent is a shorter dwell; Section VI reports what happens instead.

Twenty-seven candidates occupy a 32-amplitude register, so the whole rule is a handful of vector operations costing microseconds inside a 33 ms frame budget. We claim no quantum speedup and simulate no hardware: the claim concerns the rule's functional form, and the amplitude encoding is a product of per-feature factors and therefore classically simulable by construction.

V. Evaluation Protocol

Each session presents 21 targets, 13 for fitting and 8 held out at positions spanning corner, edge and centre, with five fixations of 300 ms retained per target after discarding the first 150 ms. Screen dimensions and viewing distance are recorded, so the conversion to degrees is measured, not assumed.

All splits are **grouped by target**. The five fixations at one position are one observation measured five times; splitting by sample leaves near-identical siblings in the training set and selects a regulariser far too small. Results are leave-one-target-out over all 21 positions, and the unit of analysis is the participant, not the frame.

The selection rule is evaluated by end-to-end typing simulation over a 16,839-word vocabulary, the gaze likelihood being the exact Gaussian integral over key rectangles at

$\sigma \in \{30, 40, 50, 60, 75\}$ px against a 100×110 px key. Three points carry the comparison. Both arms are driven from one seeded generator, so any difference is the rule alone. Because a prompt always yields the correct character, any rule reaches $100\,\%$ accuracy by prompting constantly, so we sweep the threshold and compare **commit accuracy at matched prompt rate**. And σ_φ is tuned on a 60-sentence training split and reported on a disjoint one, because a gain that does not survive the split is not a gain.

VI. Results

A. The front end: canthus normalisation

Canthus normalisation is the paper's largest effect. Table II puts held-out error at $5.552 \pm 0.488°$ for the deployed eyelid-relative features and $3.762 \pm 0.438°$ with the canthus components added, a reduction of $1.791°$ or $32.3\,\%$. The comparison is paired within participant over eleven sessions from nine participants: $t(8) = 9.31$, $p = 1.4 \times 10^{-5}$, Wilcoxon $W = 0$, $p = 0.0039$, $d_z = 3.10$, with a $95\,\%$ confidence interval on the reduction of $[1.35, 2.23]°$. *Every one of the nine participants improved*, by between $19.5\,\%$ and $44.6\,\%$; Fig. 3 overlays all nine.

Three rows isolate the cause. An iris-ring radius cue yields $0.4\,\%$, well within noise, so the gain is attributable to the canthi and not merely to having more features. Head pose alone is worth $5.7\,\%$ — the size of the effect the canthus frame subsumes — yet added *on top of* that frame it yields nothing detectable ($p = 0.439$), because the canthus frame is already head-attached and an explicit pose stage re-derives information the representation has removed. The same holds for radius ($p = 0.071$) and both together ($p = 0.112$), while removing canthus normalisation costs 40–47 % at $p < 0.001$. It is therefore the minimal sufficient addition, not one useful feature among several.

Roll invariance holds numerically as well as analytically: rotating a synthetic face through 0–40°, the canthus components drift by 4×10^{-16} to 2×10^{-15} while an image-axis formulation of the same offset drifts by 0.069. We report this as a property of the representation rather than a measured gain, because the achieved roll across our sessions never exceeds $6.5°$ — participants tilt far less than they are asked to.

B. The selection rule at matched prompt rate

The interference channel does not pay. Tuned on the training split, the best phase scale was $\sigma_\varphi = 0.40$, worth $+0.03$ percentage points there. On held-out sentences that advantage disappears: across twenty $(\sigma, \text{prompt rate})$ cells, four of which Table III shows, the median change is -0.03 percentage points, with zero significant wins and zero losses. The two posteriors do differ, just negligibly and without direction — the mean Kullback–Leibler divergence is 4.2×10^{-4} nats per keystroke — so the null is not an artefact of a rule that fails to engage. Nor is it an artefact of too small a scale: scales above 0.6 produce substantial interference and *lose*, falling from $97.0\,\%$ to $95.3\,\%$ on the training split as σ_φ rises from 0 to 1.0.

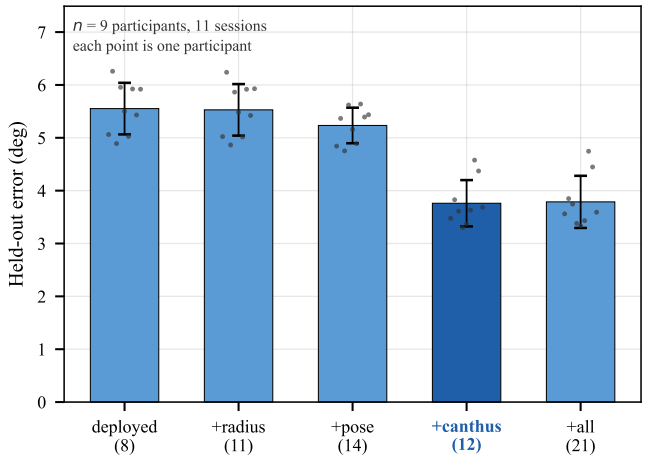


Fig. 3. Held-out error by feature set. Bars are the mean across the nine participants and whiskers one standard deviation; each participant contributes the mean of their own sessions, and all nine are overlaid as points so the spread is visible rather than summarised. Every participant improves under canthus normalisation, and adding iris radius or head pose on top of it changes nothing detectable.

TABLE II
Feature Ablation. Leave-One-Target-Out Over 21 Positions, Eleven Sessions From Nine Participants. Lower is Better.

Feature set	Dim	Error (deg)	vs deployed
Deployed (eyelid-relative)	8	5.552 ± 0.488	baseline
+ iris-ring radius	11	5.529 ± 0.488	$+0.4\,\%$
+ head pose only	14	5.233 ± 0.337	$+5.7\,\%$
+ canthus	**12**	$\mathbf{3.762 \pm 0.438}$	$\mathbf{+32.3\,\%}$
+ canthus + pose	21	3.788 ± 0.493	$+31.8\,\%$
+ all of the above	24	3.663 ± 0.474	$+34.0\,\%$

Mean $\pm$ standard deviation over the nine participants, each contributing the mean of their sessions. Adding radius, pose or both on top of the canthus frame changes nothing detectable ($p = 0.071$, 0.439, 0.112).

Amplitude amplification does not shorten dwell either. Applied after the conditioning unitary it is a no-op by construction: the conditioned state already places 94–98 % of its amplitude on consistent letters, so the optimal round count evaluates to zero and the operator never fires. The quadratic speedup is real but addresses a bottleneck this system does not have, because the admissible subspace is not rare. Forced to act on the raw gaze register, mean dwell rises from 2.63 to 2.80 frames and commit accuracy falls from $98.4\,\%$ to $96.4\,\%$ ($p = 0.48$, paired over the 249 keystrokes both arms committed).

C. Why interference fails: a sign, not a magnitude

A null result is more useful with a mechanism attached, and this one is sharp. For interference to help, the coherence penalty must fall more heavily on wrong letters than on right ones. Fig. 4 measures it, and it is negative at every phase scale, worsening monotonically from -0.004 at $\sigma_\varphi = 0.2$ to -0.108

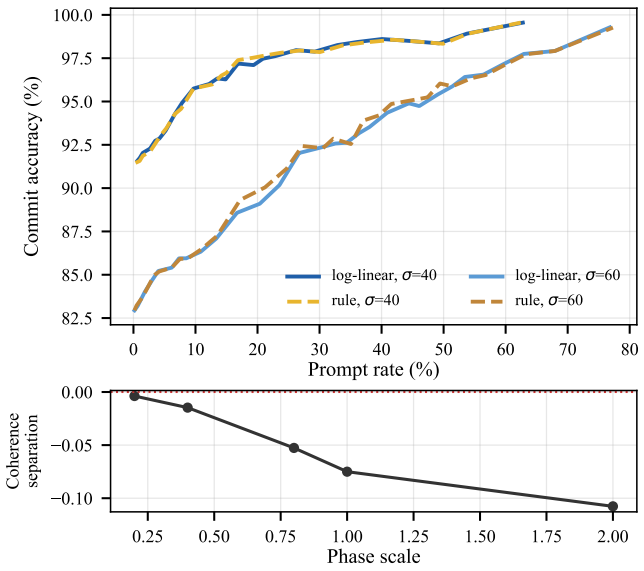


Fig. 4. Top: the selection rule traces the same accuracy/prompt-rate frontier as the fusion it generalises, at both noise levels. Bottom: the reason. The coherence separation between correct and incorrect letters is negative at every phase scale and falls monotonically, so the interference penalty lands on the correct letter rather than on the wrong ones.

TABLE III
SELECTION RULE AGAINST DEPLOYED LOG-LINEAR FUSION, HELD-OUT, AT MATCHED PROMPT RATE. EACH CELL GIVES THE BASELINE COMMIT ACCURACY AND THE RULE'S CHANGE.

σ (px)	**2 % prompts**	**10 %**	**20 %**
30	0.971 +0.003	0.986 −0.001	0.988 −0.000
40	0.921 −0.002	0.958 −0.000	0.973 +0.003
50	0.888 −0.002	0.914 −0.004	0.945 −0.001
60	0.841 −0.000	0.862 +0.001	0.891 +0.008
75	0.778 −0.000	0.819 +0.001	0.851 −0.004

Median change over all twenty $(\sigma, \text{prompt rate})$ cells -0.0003; significant wins **0**, losses **0**, by paired Wilcoxon over sentences.

at $\sigma_\varphi = 2.0$. The upper panel shows the consequence: the two frontiers lie on top of one another.

The cause is structural. The correct letter following an English prefix is typically the one with the *richest* vocabulary support: many genuine continuations, therefore many hypotheses, therefore the widest spread of phases and the lowest coherence. A wrong letter is typically backed by one obscure completion and sits at coherence 1. Any phase computed from hypothesis likelihood thus penalises the letter with the strongest linguistic evidence. We rejected two alternative explanations: raising the hypothesis budget from 4 to 4096 moves mean coherence only from 0.982 to 0.988, and flattening the within-letter weights to uniform changes it from 0.9883 to 0.9879. The limitation is the phase *source*. A language model offers no feature orthogonal to the likelihood that already sets the magnitude, so the phase channel carries no independent information for interference to exploit.

VII. LIMITATIONS

Nine participants from one institution are not a sample of the population that would use an assistive keyboard, so $[1.35, 2.23]°$ describes this group rather than eye geometry in general. Four participants contributed two sessions and five contributed one, so between-session variance rests on four people.

The eyelid mechanism is supported but not demonstrated causally. The decisive experiment holds the gaze target fixed while perturbing eyelid aperture, and the prediction is specific: eyelid-relative features should degrade sharply and canthus-relative features should not.

The selection-rule results come from simulation, with isotropic Gaussian noise standing in for the anisotropic, correlated error a real tracker produces. The null is specific to one vocabulary and one language — our claim is that likelihood-derived phase is mis-signed for English letter prediction, not that interference-based fusion is useless — and since coherence is the first quantity a noisy device degrades, a null obtained in exact simulation is if anything optimistic.

Populations this system does not serve. Two exclusions are boundaries of applicability rather than shortfalls in accuracy, and we state them because an assistive technology that overstates its reach does harm. The method assumes a commanded fixation is held, so *oculomotor instability* breaks it by design: nystagmus leaves no stable fixation to average, strabismus means a binocular iris average corresponds to no single gaze direction, and saccadic intrusions defeat the dwell centroid. This is pointed, because motor-impaired users are more likely to present with these conditions, and a scanning interface remains the appropriate design there. *Corrective eyewear* refracts the apparent iris position, reflects under overhead light, and occludes the lateral canthus at moderate yaw — removing the very landmark (3) is built on. We did not record spectacle wearers, and the accuracy reported here should not be expected to hold for them.

VIII. CONCLUSION

We set out to improve how a gaze keyboard decides, and improved how it estimates instead. The quantum conditional selection rule is a strict generalisation of log-linear fusion: it recovers the incumbent exactly at zero phase, and its interference cross-terms are provably beyond any classical mixture of the same hypotheses. Measured on identical noise at matched prompt rate it is worth a median -0.03 percentage points, and amplitude amplification lengthens dwell rather than shortening it.

The explanation is the more durable contribution. Interference fails here for a reason no retuning addresses: the correct letter after an English prefix has the richest vocabulary support, hence the most hypotheses, the widest phase spread and the lowest coherence. Phase derived from hypothesis likelihood penalises exactly the evidence it should reward. For interference to be useful, phase must come from a source orthogonal to the magnitude, and a language model does not provide one.

The accuracy came from the front end. Measuring a continuous coordinate in degrees, rather than as region classification, exposed a defect in a step the literature treats as settled: normalising the iris offset by the eyelid aperture couples the vertical channel to a structure that moves independently of gaze. Normalising against the canthi instead cuts held-out error by $32.3\,\%$ across nine participants, improves every one of them, explains an axis asymmetry reported but undiagnosed in prior work, and makes an explicit head-pose stage unnecessary — for four features and no additional runtime cost.

ACKNOWLEDGMENT

The authors gratefully acknowledge the National Institute of Technology Tiruchirappalli for providing access to its laboratory facilities, under which this work was initiated and carried out to completion. The authors also thank the Department of Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies, for its continued support.

REFERENCES

[1] B. Waideman and P. T. Aquino Junior, "Augmentative and alternative communication using eye tracking and word recommendation using language models," *IEEE Latin America Transactions*, vol. 23, no. 8, pp. 637–645, Aug. 2025.

[2] M. R. Islam, M. S. Rahman, and A. Sarkar, "Computer vision based eye gaze controlled virtual keyboard for people with quadriplegia," in *Proc. Int. Conf. Automation, Control and Mechatronics for Industry 4.0 (ACMI)*, Rajshahi, Bangladesh, Jul. 2021, pp. 1–6.

[3] V. Abhaya, A. S. Bharadwaj, C. C. Bagan, K. Dhanraj, and G. Shyamala, "Eye-Move: An eye gaze typing application with OpenCV and Dlib library," in *Proc. Int. Conf. Automation, Computing and Renewable Systems (ICACRS)*, Dec. 2022, pp. 952–957.

[4] S. T. S. Rafid, A. J. Ajwad, and A. K. M. A. M. Azad, "Empowering quadriplegic patients: A computer vision based approach for keyboard control using eye gaze and blink," in *Proc. 26th Int. Conf. Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, Dec. 2023, pp. 1–5.

[5] B. G. Varshini, A. Manimegalai, S. D. Abirami, and S. Kalaiselvi, "Eye-gaze controlled virtual keyboard using blink detection for accessible hands-free communication," in *Proc. Int. Conf. Computational, Communication and Information Technology (ICCCIT)*, Feb. 2025, pp. 49–56.

[6] A. SK, E. D. A, H. B, and H. M, "Eye gaze tracking for virtual mouse control using computer vision," in *Proc. 4th Int. Conf. Artificial Intelligence and Machine Learning Applications (AIMLA)*, Apr. 2026, pp. 1–6.

[7] K. Kathpal, S. Negi, and S. Sharma, "iChat: Interactive eyes for specially challenged people using OpenCV Python," in *Proc. 9th Int. Conf. Reliability, Infocom Technologies and Optimization (ICRITO)*, Noida, India, Sep. 2021, pp. 1–5.

[8] V. I. Saraswati, R. Sigit, and T. Harsono, "Eye gaze system to operate virtual keyboard," in *Proc. Int. Electronics Symp. (IES)*, Denpasar, Indonesia, Sep. 2016, pp. 175–179.

[9] D. A. Padilla, J. A. B. Adriano, J. R. Balbin, I. G. Matala, J. J. R. Nicolas, and S. R. R. Villadelgado, "Implementation of eye gaze tracking technique on FPGA-based on-screen keyboard system using Verilog and MATLAB," in *Proc. IEEE Region 10 Conf. (TENCON)*, Penang, Malaysia, Nov. 2017, pp. 2771–2776.

[10] D. Li, L. Zhang, W. Hu, H. Li, and X. Zhou, "Dwell time adaptive three-way text entry by gaze," in *Proc. 31st Chinese Control and Decision Conf. (CCDC)*, Nanjing, China, Jun. 2019, pp. 5295–5300.

[11] C. Lugaresi *et al.*, "MediaPipe: A framework for building perception pipelines," 2019, arXiv:1906.08172.

[12] G. Casiez, N. Roussel, and D. Vogel, "The one Euro filter: A simple speed-based low-pass filter for noisy input in interactive systems," in *Proc. ACM CHI*, 2012, pp. 2527–2530.

[13] G. Brassard, P. Høyer, M. Mosca, and A. Tapp, "Quantum amplitude amplification and estimation," in *Quantum Computation and Information*, AMS Contemporary Mathematics, vol. 305, 2002, pp. 53–74.

[14] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proc. 28th Annu. ACM Symp. Theory of Computing (STOC)*, 1996, pp. 212–219.

[15] P. O. Kristensson and S. Zhai, "SHARK^2: A large vocabulary shorthand writing system for pen-based computers," in *Proc. ACM UIST*, 2004, pp. 43–52.